

JAGADEESH REDDY VANGA

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Education

Stony Brook University, Stony Brook, New York

Jan 2023 – Dec 2024

Master of Computer Science

KL University, Vijayawada, India

Jun 2016 – May 2020

Bachelor of Technology in Computer Science and Engineering

Experience

Skit.ai - Software Engineer(Solutions and Machine Learning)

Oct 2021 – Dec 2022

- Designed and implemented bespoke voice bots for four enterprise fin-tech clients, automating 10% (100,000/day) of customer service calls.
- Facilitated coordination among cross-functional teams, leading to successful delivery of 20+ high-priority features.
- Analyzed client call data using SQL-based pipelines and dashboards, uncovering trends in caller behavior leading to informed product optimizations and customer engagement strategies.
- Streamlined technical onboarding process for hybrid roles by standardizing training modules and documentation, reducing ramp-up time to 7 days.
- Optimized voice bot solutions by fine-tuning XLMR models for multilingual support and generating domain-specific labeled training data, increasing response accuracy by 12% and scaling for peak loads by 20%.

Eunimart MultiChannel - Team Lead AI Development

Jun 2019 – Sept 2021

- Directed technical efforts across 8 diverse projects, managing timelines, delegating tasks, and cultivating effective communication within a 10-member team, achieving a 100% project success rate.
- Upgraded legacy AI systems using modern architectures and optimization techniques, resulting in a 20% increase in throughput and a 12% improvement in accuracy.
- Developed a Similar Products Predictor, deployed an ensemble model of VGG16 and BERT, identifying top 10 similar products based on images and product details, boosting customer recommendations by 15%, and deployed via Flask.
- Designed a Sales Prediction model based on XGBoost and Facebook's Prophet, improving forecast accuracy by 15%, streamlined inventory management and demand planning for enhanced operational efficiency.
- Implemented an Image Optimizer service featuring blur removal, background elimination, and resolution upgrades using OpenCV, reducing professional photo shoot dependency by 40% while enhancing listing quality.
- Architected a robust web scraper for e-commerce platforms (Amazon, eBay, Lazada, Flipkart), building a database of over 50 million products, with a 30% improvement in scraping efficiency and data quality.

Academic Projects

DataPulse - Customer Segmentation Utilizing AI:

(GitHub Repository)

- Devised an AI-driven Customer Segmentation system leveraging clustering and PCA, predicted similar products via Word2Vec, and boosted sales forecasting accuracy with Facebook's Prophet. Integrated Google's Gemma locally for interactive dashboard insights, earning Graduate Best in Show at Hack@CEWIT 2024.

TeleMedi:

(GitHub Repository)

- Engineered a comprehensive telemedicine platform for HackPrinceton Fall 2024, enabling remote consultations with features including user authentication, profile management, appointment scheduling, doctor recommendations, and emergency alerts using React for frontend, Flask for backend, and JWT for secure authentication.

Adversarial Testing for Robust Content Moderation - Research Project:

(GitHub Repository)

- Conducted a systematic analysis of NSFW detection systems on social media platforms, leveraging black-box attack methodologies. Formulated a patch attack using GRAD-CAM, bypassed content moderation filters with 93% accuracy, uncovering critical vulnerabilities and providing actionable recommendations to improve robustness.

Technical Skills

Programming Languages: Python (advanced), Java (intermediate), SQL (advanced), C, JavaScript, Shell Scripting

Machine Learning and Data Science: PyTorch, TensorFlow, Scikit-learn, OpenCV, XGBoost, Pandas, NumPy

Web Development: Flask, Django, ReactJS, RESTful APIs, Web Scraping

Database Technologies: Postgres, MySQL, MongoDB, DB2, DynamoDB, RDS, Redis

Cloud Technologies: AWS, GCP, Azure

Version Control and CI/CD: GIT, GitHub, GitLab, Bitbucket, Jenkins, Argo WorkFlows)

Tools and IDEs: VS Code, Eclipse, Jira, R Studio

Software Engineering Concepts: Object-Oriented Programming, Software Design Patterns, System Implementation, Agile Development